

SCOPE · B.TECH PROJECT

# Adaptive Physics-Informed Battery Digital Twin for SOH and RUL Prediction

*Energy AI · Battery Management Systems · Physics-Informed Neural Networks*

## TEAM

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SOH · RUL



# Project Overview & Use Cases

DOMAIN

Energy AI · Battery Management Systems

TYPE OF DNN

Physics-Informed Neural Network (PINN) + Feedforward DNN

THE CHALLENGE

Li-ion batteries degrade with cycling, temperature and age. Pure deep-learning models need large labeled datasets and can be physically inconsistent; pure physics models are too slow for real-time use.

OUR APPROACH

An Adaptive PINN Battery Digital Twin fuses PINNs, a Transformer Encoder and Incremental Adaptive Learning to continuously estimate SOH and RUL — adapting to new data without full retraining.

WHERE IT APPLIES

| Use Case                  | Impact                                                                        |
|---------------------------|-------------------------------------------------------------------------------|
| Electric Vehicles         | Tesla, BYD, Hyundai, Tata EV, MG — predict battery replacement before failure |
| Battery Swapping Stations | Identify which battery packs should be retired                                |
| Solar Energy Storage      | Monitor home batteries and predict maintenance                                |
| Grid-Scale Storage        | Avoid sudden, unplanned battery failures                                      |
| Drone Batteries           | Estimate safe remaining flying time                                           |
| Space Missions            | Battery swaps are impossible — lifetime prediction is critical                |
| Smartphones & Laptops     | Sharper degradation estimates than today's health meters                      |

# Problem Statement

Lithium-ion batteries power electric vehicles and renewable energy storage, but their performance degrades with charge cycles, temperature, current load and age. Accurately estimating State of Health (SOH) and Remaining Useful Life (RUL) is essential for safety, reliability and efficient maintenance.



## Deep Learning Models

- Require large volumes of labeled training data
- May produce physically inconsistent predictions
- Limited generalization to unseen operating conditions

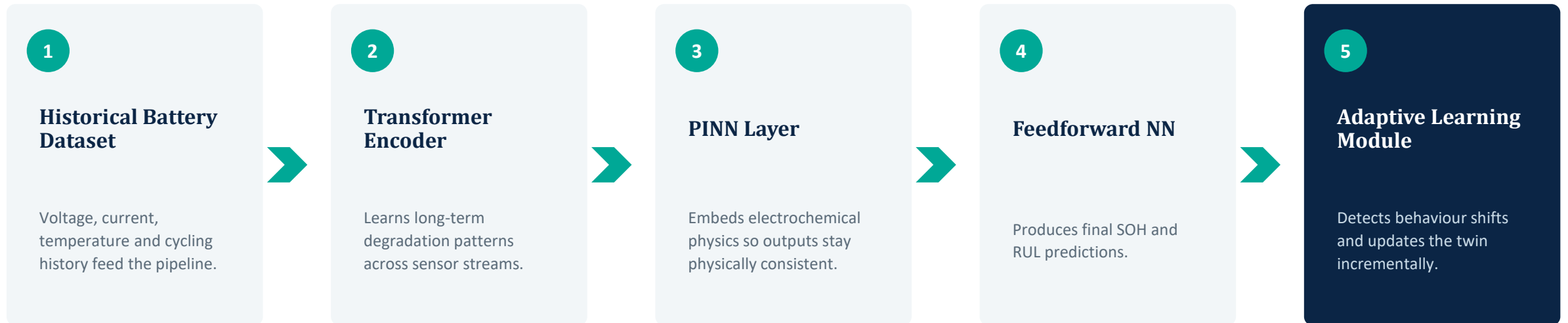


## Physics-Based Models

- Computationally expensive to run
- Difficult to deploy for real-time estimation
- Rely on complex electrochemical simulations

# Proposed Approach & Architecture

Real-time sensor data, electrochemical battery physics and deep learning combine to estimate SOH and RUL. The Digital Twin continuously adapts to new battery behaviour through incremental learning — no full retraining required.



# Base Paper

## *Physics-Informed Neural Networks for Battery Modeling and State Estimation: A Methodological Review*

Renewable and Sustainable Energy Reviews, 2026

### WHY THIS PAPER WAS SELECTED

- Comprehensive review of Physics-Informed Neural Networks (PINNs) for battery applications
- Covers SOH, RUL, SOC, Digital Twins, thermal diagnosis and internal state reconstruction
- Explains multiple methods for embedding battery physics into neural networks
- Discusses training stability, domain generalization, uncertainty quantification and interpretability
- Lays out future research directions for intelligent Battery Management Systems

# Literature Review & Research Gap

RESEARCH GAP

Existing PINN models are trained offline, adapt poorly to changing battery behaviour, and remain difficult to deploy in real-time BMS. Lightweight, continual-learning methods are still an open problem.

| Paper                                | Key Contribution                                                                | Limitation                                                   | Our Improvement                                                                  |
|--------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------|
| Applied Energy (2025)                | PINN + Transformer + multi-task learning predicts SOH, RUL and degradation path | Offline training; adapting to new behaviour needs retraining | Adaptive Digital Twin with incremental fine-tuning                               |
| Energy (2026)                        | EM-PINN integrates SEI growth physics for accurate SOH estimation               | SOH-only; no Digital Twin or continual adaptation            | Combines PINN with a Digital Twin predicting SOH and RUL                         |
| Scientific Reports (2025)            | Cycle-aware physics-informed framework for RUL prediction                       | Offline; limited personalization per battery                 | Continuously updates predictions with new operational data                       |
| Complex & Intelligent Systems (2025) | PINN + contrastive learning for low-label SOH estimation                        | Train-then-deploy workflow; no post-deployment adaptation    | Supports incremental learning without full retraining                            |
| Proposed Work                        | Transformer + PINN + Feedforward DNN + Digital Twin + Adaptive Learning         | —                                                            | Incremental fine-tuning on new battery behaviour, preserving physics constraints |

# Datasets Used & Rationale

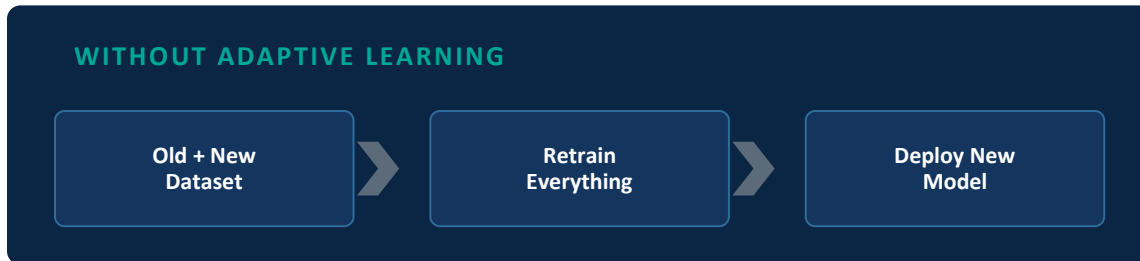
SELECTED DATASETS

Public benchmark battery datasets cited across the reviewed papers, spanning lab-aged, second-life, and vehicle-field conditions, chosen to validate SOH/RUL estimation under diverse chemistries and operating regimes.

| Dataset                        | Scope                                                                                    | Link                                                                                                                          | Why This Dataset                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| NASA & Oxford Battery          | Lab-aged Li-ion cells; narrow current/temperature range; classic PINN-SOH benchmark      | <a href="https://data.nasa.gov/PCoE/">data.nasa.gov (PCoE)</a> / <a href="#">howey-lab Oxford Battery Degradation Dataset</a> | [0] Standard reference set used by our base-paper PINN methodology review to validate degradation-physics constraints |
| Stanford-MIT-Toyota (Severson) | 124 LFP/graphite cells, 72 fast-charging policies, cycled to end-of-life                 | <a href="https://data.matr.io/">data.matr.io</a> (Battery Evaluation and Early Prediction project)                            | [1] Used by the RUL-focused papers for early-cycle degradation trajectories and fast-charging diversity               |
| TJU & XJTU                     | Multi-chemistry aging data with IC-curve health features across varied cycling protocols | <a href="#">Tianjin University</a> / <a href="#">Xi'an Jiaotong University</a> battery aging repositories                     | [2] Referenced by the time-series PINN-SOH literature to test cross-dataset transfer of the Adaptive Learning module  |
| CALCE & Panasonic 18650PF      | Dynamic charge/discharge cycles, -10°C to 20°C, multi-temperature conditions             | <a href="https://calce.umd.edu/">calce.umd.edu</a> (Battery Research Group) / <a href="#">NASA Ames Prognostics Center</a>    | [3] Provides low-temperature, condition-diverse cycles to stress-test physics-informed robustness                     |
| Second-Life LIB (SL-LIB)       | 84 retired batteries, 21 working conditions, real second-life usage patterns             | Self-designed dataset (Bayesian PINN, Applied Energy 2025)                                                                    | [4] Real-world retired-battery data supports validating adaptive learning on genuinely shifting degradation behaviour |

# Novel Contribution: Adaptive Incremental Learning

Traditional models are trained once on historical data. When battery behaviour shifts with aging, temperature or charging patterns, the whole model must often be retrained — expensive and impractical for real-time BMS.



## HOW IT WORKS

- 1 Monitor real-time sensor data (V, I, temperature, SOC, cycle count)
- 2 Detect changes in degradation behaviour
- 3 Store important new data in a replay buffer
- 4 Incrementally fine-tune the existing model
- 5 Update the Digital Twin — no full retraining

## ADVANTAGES

- Eliminates full model retraining
- Adapts to changing battery behaviour
- Personalized prediction per battery
- Physics-consistent predictions via PINN
- Fit for real-time BMS deployment

# Applications

1

## Electric Vehicles

Predict battery replacement before failure occurs.

2

## Solar Storage

Monitor home batteries and predict maintenance needs.

3

## Smart Grids

Avoid sudden, costly battery failures at scale.

4

## Drones

Estimate safe remaining flying time in-flight.

5

## Laptops & Smartphones

More accurate battery degradation estimates.

## CONCLUSION

# Project Summary

- Proposed an Adaptive Physics-Informed Battery Digital Twin for accurate SOH and RUL estimation.
- Integrated Transformer Encoder, PINN and Feedforward DNN to combine data-driven learning with battery physics.
- Introduced incremental adaptive learning to keep the Digital Twin current without full model retraining.
- Overcomes key limits of pure deep-learning and pure physics-based models with real-time, physically consistent predictions.

**Potential impact:** Electric Vehicles, Battery Management Systems, Renewable Energy Storage, Smart Grids, Drones and Consumer Electronics.

# References

1

**Physics-informed neural networks for battery modeling and state estimation: A methodological review**

*Renewable and Sustainable Energy Reviews*, 2026 <https://www.sciencedirect.com/science/article/pii/S1364032126005253>

2

**Physics-informed neural network for co-estimation of state of health, remaining useful life, and short-term degradation path in Lithium-ion batteries**

*Applied Energy*, 2025 <https://www.sciencedirect.com/science/article/pii/S0306261925011572>

3

**Physical informed neural network for SOH estimation of Lithium-ion battery with electrochemical mechanism**

*Energy*, 2026 <https://www.sciencedirect.com/science/article/pii/S0360544225053964>

4

**A cycle-aware and physics-informed framework for battery remaining useful life prediction**

*Scientific Reports*, 2025 [https://www.researchgate.net/publication/399255851\\_A\\_cycle-aware\\_and\\_physics-informed\\_framework\\_for\\_battery\\_remaining\\_useful\\_life\\_prediction](https://www.researchgate.net/publication/399255851_A_cycle-aware_and_physics-informed_framework_for_battery_remaining_useful_life_prediction)

5

**Physics-informed neural network and momentum contrastive learning for battery state of health estimation**

*Complex & Intelligent Systems*, 2025 [https://www.researchgate.net/publication/398786756\\_Physics-informed\\_neural\\_network\\_and\\_momentum\\_contrastive\\_learning\\_for\\_battery\\_state\\_of\\_health\\_estimation](https://www.researchgate.net/publication/398786756_Physics-informed_neural_network_and_momentum_contrastive_learning_for_battery_state_of_health_estimation)

# Model Design & Experimental Study

Completed implementation, experimental evaluation, and benchmark analysis

## NASA-first evidence bundle

54 hyperparameter combinations · 3 random seeds · 18 ablation rows · 12 robustness rows

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## Key Evaluation Focus

Model architecture · Multiple models · Hyperparameter tuning · Experimental analysis · Methodology & Development Summary

# Phase II focus: DL model design and hyperparameter tuning

The required work is explicit: design the DL models, tune their hyperparameters, deploy multiple models, and compare results.



## Data contract

NASA B0005, B0006, B0007, and B0018 are parsed into ordered discharge-cycle records with voltage, current, temperature, time-gap, and capacity.



## Model stack

DNN, GRU, Transformer-only, PINN, and adaptive Transformer-plus-physics models run under the same held-out-cell protocol.



## Online learning

A drift detector triggers low-cost replay-buffer updates as the held-out stream arrives in time order.



## Experimental Evidence

The run exports final comparison tables, ablations, robustness results, physical diagnostics, figures, and a reflection script.



## Scope & Boundary

SEI-PDE physics and broad multi-chemistry validation remain extensions because the NASA POC does not expose every required electrochemical variable.



## Experimental Rigor

Reported paper metrics stay separate from new project metrics. The project claims only what the executed protocol measures.

# NASA cells and a leakage-safe evaluation protocol

The official NASA PCoE repository provides the first committed dataset for this proof of concept.

| Item     | Project choice                                                | Why it matters                                         |
|----------|---------------------------------------------------------------|--------------------------------------------------------|
| Cells    | B0005 / B0006 / B0007 train · B0018 held out                  | Battery-level separation tests transfer to a new cell. |
| Labels   | SOH = capacity / nominal capacity × 100 · EOL = 70% SOH       | RUL = EOL cycle – current cycle.                       |
| Inputs   | Voltage · current · temperature · delta_t · cycle index       | Measured timing is retained with masks.                |
| Protocol | 504 train windows · 132 test windows · 33 initial · 99 stream | Later cycles arrive sequentially for adaptation.       |
| Seeds    | 7 · 17 · 27                                                   | Final tables report mean ± standard deviation.         |

EOL RULE

70% SOH

HELD-OUT CELL

B0018

CYCLES PARSED

636 total

ORDER

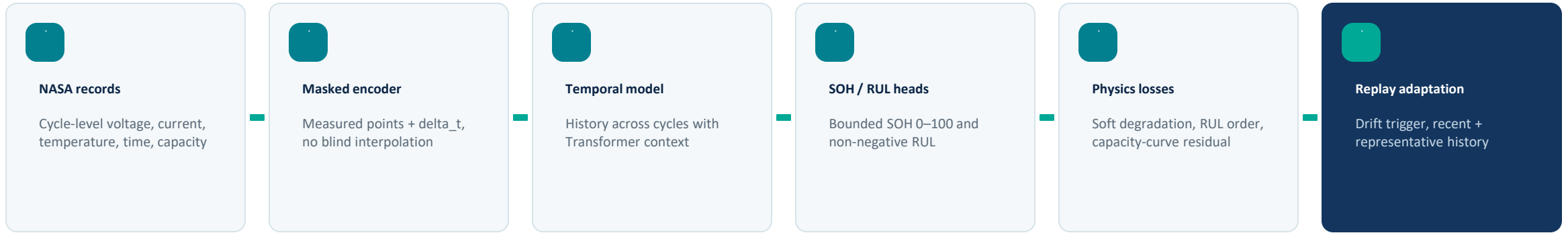
Time-aware

METRIC UNIT

RUL in cycles

# From measured cycle to adaptive digital-twin state

The first-stage physics model uses relationships supported by the NASA measurements.



Training objective

$$L_{\text{total}} = L_{\text{data}} + \lambda_{\text{phys}} L_{\text{phys}} + \lambda_{\text{cons}} L_{\text{cons}}$$

The SEI-growth PDE formulation remains a Stage 2 extension until every variable and parameter is available and verified.

# The selected configuration came from the complete grid

Every requested combination ran across seeds 7, 17, and 27 before the final evidence pass.

| Grid dimension     | Values tested           |
|--------------------|-------------------------|
| Hidden size        | 64 · 128                |
| Learning rate      | 0.0001 · 0.0005 · 0.001 |
| Physics weight     | 0.1 · 1 · 10            |
| Replay size        | 128 · 512 · 2048        |
| Total combinations | 54 × 3 seeds = 162 runs |

Selected configuration

**Hidden size 64**  
**Learning rate 0.001**  
**Physics weight 0.1**  
**Replay size 128**

The final evidence runner then used the selected configuration on the full default cycle-window representation.

GRID STATUS

54 / 54 complete

MODEL COUNT

5 comparisons

EVIDENCE MODE

Held-out cell

OUTPUT

JSON + CSV + PNG

# Online replay adaptation delivered the best SOH accuracy

All five models trained under one identical protocol (six epochs each, uniform, no model favored); single seeded run.

| Model                             | SOH MAE (%)  | RUL MAE (cycles) | Online adaptation  |
|-----------------------------------|--------------|------------------|--------------------|
| DNN                               | 7.39         | 31.7             | No                 |
| GRU                               | 7.03         | 29.4             | No                 |
| Transformer-only                  | 10.09        | 37.3             | No                 |
| PINN without adaptation           | 7.75         | 41.9             | No                 |
| Adaptive twin (frozen -> adapted) | 7.64 -> 5.43 | 38.9             | Yes (CUSUM+replay) |

Adaptive twin (ours)

7.64% frozen -> 5.43% adapted SOH MAE (28.8% relative improvement) - 38.9 RUL MAE

Best frozen-model SOH row

GRU: 7.03% SOH MAE (frozen, no adaptation available)

# What to tell faculty: the headline result

All five models: same data split, same optimizer, same 6 epochs. Only the Adaptive Twin runs online replay adaptation - that is the proposed contribution.

## What the run supports



### Adaptation benefit

SOH MAE 7.64% (frozen) to 5.43% after 8 drift-triggered updates; lowest of all five models.



### Low update cost

22.9% of full-retrain compute, about 1.2 minutes, no retraining from scratch.

## Honest limits (say these before asked)

- GRU has better RUL MAE (29.4 vs 38.9 cycles)
- Single seeded run; 4 NASA cells only
- Soft monotonic penalty, not perfect physics
- Not a ranking against the published papers

## Compared with

DNN 7.39% · GRU 7.03% · PINN (offline) 7.75% · Transformer 10.09% SOH MAE on held-out cell B0018.

# Data preparation: from NASA .mat files to model input

Every step is in the notebook; irregular timing kept as `delta_t` and masks, and no future cycles enter the inputs.

## Parsing and labels



### Parse and label

Keep discharge cycles only: 636 cycles (B0005/B0006/B0007: 168 each, B0018: 132). SOH =  $100 \times C / 2.0 \text{ Ah}$ .



### RUL rule

EOL = first cycle with SOH at or below 70%; RUL =  $k_{\text{EOL}} - k$ , fixed before any windowing.

## Leakage safety and input encoding

- Cell-aware split: train B0005/6/7, test B0018
- Input per point: [V, I, T, `delta_t`, bias channel]
- Labels scaled: SOH / 100, RUL / 250
- 12-cycle windows, up to 256 points, padded + masked
- 504 train windows / 132 test windows

## Why it matters

A random row split would leak later cycles of the same battery; holding out a whole cell tests real generalization.

# Data analysis: what the NASA cells look like

EDA figure (eda\_summary.png) and dataset audit are the first cells of the notebook.

## Dataset audit



### Capacity fade differs per cell

Capacity spans 1.15-2.04 Ah; SOH ends near 64% (B0005), 58% (B0006), 70% (B0007), 67% (B0018).



### Different lifetimes

Max RUL 124 / 108 / 167 / 96 cycles, so RUL is harder than SOH across cells.

## What the analysis told us to do

- Capacity-regeneration spikes exist (B0006 starts above 100%)
- So monotonic degradation is a soft penalty only
- Points per cycle vary (about 180-370): irregular time
- So cycles are padded and masked, not resampled

## Takeaway

EDA drove three design choices: soft monotonic loss, masked irregular timing, and a held-out-cell protocol.

# How the online adaptation works

The stream of B0018 is revealed cycle by cycle; the model predicts first, then may update.

## Mechanism



### Drift detection (CUSUM)

Accumulates the SOH residual; triggers when the sum passes the threshold ( $k = 0.03$ ,  $h = 0.35$ ). 8 triggers in this run.



### Replay fine-tune

128-sample replay buffer; each trigger fine-tunes on 96 samples for 6 epochs at  $lr = 4e-4$  with the physics loss kept on.

## Where to look in the notebook

- File:  
`Adaptive_Physics_Informed_Battery_Digital_Twin_executed.ipynb`
- Data audit + EDA: first code cells
- Models and physics losses: middle cells
- Results, trajectory plot, efficiency: last cells

## Evidence

Trajectory plot: the frozen model stays flat while the adapted twin follows the true SOH decline after each drift event.

# Replay adaptation delivered the low-cost online update

The ablation table shows which components mattered in the ordered B0018 stream. (Source: earlier 3-seed battery\_twin evidence bundle - different protocol from the notebook comparison.)

| Variant            | Post-shift SOH MAE | Improvement | Work    | Events |
|--------------------|--------------------|-------------|---------|--------|
| Proposed adaptive  | 1.74               | 80.63%      | 8.75%   | 7      |
| No physics         | 1.23               | 86.98%      | 8.75%   | 7      |
| No Transformer     | 4.09               | 50.39%      | 8.75%   | 7      |
| No replay          | 3.97               | 53.23%      | 8.75%   | 7      |
| No drift detection | 1.59               | 81.91%      | 15.00%  | 12     |
| Full retraining    | 1.58               | Reference   | 100.00% | —      |

TARGET 1

80.63% improvement

TARGET 2

8.75% work

TARGET 3

–10.20% forgetting

DETECTOR

7 updates / seed

# The adaptive state remained useful under sensor corruption

Robustness scenarios use the same trained proposal and corrupt the ordered stream before adaptation.

| Scenario                   | Frozen SOH MAE | Adaptive SOH MAE | Updates |
|----------------------------|----------------|------------------|---------|
| Clean ordered stream       | 8.81           | 1.73             | 7       |
| Voltage/current/temp noise | 9.02           | 1.52             | 7       |
| 10% missing samples        | 8.78           | 1.51             | 7       |
| Sensor drift               | 8.88           | 1.69             | 7       |

| Diagnostic                    | Mean ± SD |
|-------------------------------|-----------|
| SOH bounds violations         | 0.00%     |
| Negative RUL                  | 0.00%     |
| SOH degradation violation     | 21.43%    |
| SOH/RUL consistency violation | 30.95%    |
| Capacity-curve residual       | 0.00573   |

Interpretation

Hard bounds stayed at zero. Soft monotonic diagnostics stayed nonzero, so the result supports a measured proof of concept with calibration limits rather than perfect electrochemical validity.

# Proof of experiment mapped to the course rubric

The deck now points to concrete artifacts a reviewer can inspect.

| Rubric criterion            | What to show                                                                    | Artifact                            |
|-----------------------------|---------------------------------------------------------------------------------|-------------------------------------|
| Model architecture design   | Data → masked encoder → temporal model → physics losses → heads → replay update | HTML report · Section 05            |
| Multiple models             | DNN, GRU, Transformer-only, PINN, adaptive twin                                 | final_summary.csv                   |
| Hyperparameter tuning       | 54 combinations across three seeds                                              | tuning/grid.json                    |
| Experimental analysis       | Ablations, robustness, physical checks, acceptance table                        | evidence_results.json + PNG figures |
| Presentation and reflection | 6-minute walkthrough and AI pair-programming reflection                         | video_reflection_script.md          |

Recording checklist

Show the design diagram · show all five models · open the 54-combination grid · show one ablation and one robustness figure · read PASS status exactly

# A tested adaptive POC with a clear next experiment

The strongest evidence is the low-cost online update. The main limitation is the small four-cell dataset. (Source: earlier 3-seed battery\_twin evidence bundle - different protocol from the notebook comparison.)

## What the run supports



### Adaptation benefit

80.63% lower post-shift SOH MAE than frozen.



### Low update cost

8.75% of full-retraining work.

## What the run does not prove

- Universal chemistry or temperature generalization
- Perfect monotonic degradation behavior
- SEI-PDE validity without additional variables
- A direct numerical ranking against the four papers

## Next experiment

Calibrate physics weights, add longer training, and repeat leave-one-cell-out validation before making a broader scientific claim.

# Video demonstration

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VIDEO LINK

**PASTE VIDEO URL HERE**

Replace this placeholder with the actual uploaded video URL before submitting.

*No external video URL was present in the supplied files, so this slide intentionally keeps the link field open.*

# Equations used in our NASA prototype

These are the target definitions and input contract that are active in the first reproducible implementation.

## STATE OF HEALTH

$$\text{SOH}_k = 100 \times C_k / C_{\text{nom}}$$

Measured discharge capacity  $C_k$  becomes the supervised health label. Capacity is not fed as a future deployment input.

## REMAINING USEFUL LIFE

$$\text{RUL}_k = k_{\text{EOL}} - k$$

RUL is measured in discharge cycles remaining from the current cycle  $k$ .

## END OF LIFE

$$k_{\text{EOL}} = \min\{k : \text{SOH}_k \leq 70\%\}$$

Primary project threshold. Every RUL result is interpreted against this declared EOL rule.

## DEPLOYED INPUT VECTOR

$$\mathbf{x}_k = [V, I, T, \Delta t, k]$$

Voltage, current, temperature, measured time-gap, and cycle index. Irregular timing is preserved with  $\Delta t$  and masks.

Status: implemented and evaluated on NASA B0005/B0006/B0007 → B0018

# The active training objective and physical diagnostics

The first-stage POC uses capacity/degradation consistency that the NASA records can support; it does not claim an SEI-PDE residual.

$$L_{\text{total}} = L_{\text{data}} + \lambda_{\text{phys}} L_{\text{phys}} + \lambda_{\text{cons}} L_{\text{cons}}$$

selected:  $\lambda_{\text{phys}} = 0.1$  ·  $\lambda_{\text{cons}} = 1$  · RUL scale = 250 cycles

## DATA LOSS

$$L_{\text{data}} = \text{mean}|\hat{\text{soh}} - \text{soh}| + \text{mean}|\hat{\text{rul}} - \text{rul}|/250$$

Joint SOH and RUL regression with valid-cycle masks.

## SOFT PHYSICS LOSS

$$L_{\text{phys}} = \text{mean ReLU}(\Delta\hat{\text{soh}}) + \text{mean ReLU}(\Delta\hat{\text{rul}})$$

Penalizes predicted SOH or RUL increases across ordered cycles; soft because real cells can show local regeneration.

## CURVE CONSISTENCY

$$L_{\text{cons}} = \text{mean}|\Delta\hat{\text{soh}} - \Delta\text{soh}| / 100$$

Capacity-curve shape residual between predicted and measured SOH changes.

## BOUNDED HEADS

$$\hat{\text{soh}} = 100 \cdot \sigma(z_1), \quad \hat{\text{rul}} = 250 \cdot \sigma(z_2)$$

Output contract: SOH  $\in [0, 100]$  and RUL  $\in [0, 250]$ . The nonnegative-consistency term is an explicit zero-valued future hook.

*Diagnostics reported: SOH bounds violation · negative RUL rate · degradation violation rate · SOH/RUL consistency violation · capacity-curve residual*

# Base paper: general PINN formulation

Transcribed from the methodological review, pp. 4–5. These are representative framework equations, not one new end-to-end model.

(1) NEURAL APPROXIMATION

$$\mathbf{u}\theta(\mathbf{x},t) = \text{NN}\theta(\mathbf{x},t)$$

A network represents a battery state variable such as concentration, voltage, temperature, or SOC.

(2) GOVERNING OPERATOR

$$\mathcal{N}[\mathbf{u}\theta](\mathbf{x},t; \lambda) = 0$$

$\mathcal{N}$  is the physical operator derived from electrochemical, ECM, aging, or thermal equations;  $\lambda$  denotes physical parameters.

(3) RESIDUAL

$$\mathbf{r}\theta(\mathbf{x},t) = \mathcal{N}[\mathbf{u}\theta](\mathbf{x},t; \lambda)$$

Automatic differentiation evaluates the physical-law residual at a point.

(4) PHYSICS LOSS

$$\mathbf{L}_{\text{phys}} = (1/N_p) \sum |\mathbf{r}\theta(\mathbf{x}_i,t_i)|^2$$

Residuals are averaged over  $N_p$  physics collocation points.

(5) COMPOSITE PINN LOSS

$$\mathbf{L}_{\text{PINN}} = w_d \mathbf{L}_{\text{data}} + w_p \mathbf{L}_{\text{phys}} + w_b \mathbf{L}_{\text{BC/IC}} + w_r \mathbf{L}_{\text{reg}}$$

Weights balance data fit, governing physics, boundary/initial conditions, and optional regularization.

**Implementation:** We specialize data loss, physical monotonicity, and capacity curve residual losses for battery health tracking.

# Base paper: electrochemical constraints

These equations are reviewed as physics sources for PINNs; they are not activated in the first NASA POC because the required internal variables are not observed.

(6) SOLID-PHASE DIFFUSION

$$\partial c_s / \partial t = (D_s / r^2) \partial / \partial r (r^2 \partial c_s / \partial r)$$

Fick-type diffusion in a spherical active-material particle;  $c_s$  is concentration and  $D_s$  is solid-phase diffusivity.

(7) PARTICLE BOUNDARY CONDITIONS

$$\partial c_s / \partial r|_0 = 0, \quad -D_s \partial c_s / \partial r|_{R_s} = J_n$$

Symmetry at the particle center and surface-flux condition at radius  $R_s$ .

(8) BUTLER–VOLMER KINETICS

$$j_n = i_0 [\exp(\alpha_a F \eta / RT) - \exp(-\alpha_c F \eta / RT)]$$

Interfacial reaction-current relation;  $i_0$  is exchange current density and  $\eta$  is overpotential.

(9) CHARGE CONSERVATION

$$\nabla \cdot \mathbf{i}_s = -a_s j_n, \quad \nabla \cdot \mathbf{i}_e = a_s j_n$$

Solid/electrolyte current-density balance with specific interfacial area  $a_s$ .

# Base paper: ECM state evolution and aging laws

These equations connect macroscopic circuit state, capacity fade, and monotonic degradation constraints.

(10) COULOMB COUNTING

$$dz/dt = -\eta I(t)/Q_n$$

$z$  is SOC,  $I$  is applied current,  $Q_n$  is nominal capacity, and  $\eta$  is Coulombic efficiency.

(11) RC POLARIZATION

$$dV_p/dt = -V_p/(R_p C_p) + I/C_p$$

First-order equivalent-circuit polarization branch.

(12) TERMINAL VOLTAGE

$$V_t = U_{OC}(z) - IR_o - V_p$$

Open-circuit voltage plus ohmic and polarization terms.

(13) CAPACITY-BASED SOH

$$SOH(k) = Q(k)/Q_o$$

The base review's capacity-health equation is the source for our percentage form  $SOH_k = 100Q_k/Q_{nom}$ .

(14–15) CAPACITY FADE

$$Q(k) = Q_o - \Delta Q(k) \quad \text{or} \quad Q(k) = Q_o - ak^b$$

General accumulated loss and a semi-empirical cycle-dependent approximation.

(16) MONOTONIC AGING

$$dQ/dk \leq 0 \quad \text{or} \quad Q(k+1) \leq Q(k)$$

The basis for soft degradation constraints; local regeneration can motivate a soft rather than hard penalty.

**Implementation:** We adopt capacity-derived SOH and soft monotonicity constraints to reflect degradation physics.

# Base paper: thermal and reaction-rate constraints

The review extends the PINN equation map to coupled thermal fields and temperature-dependent kinetics.

(17) LUMPED THERMAL MODEL

$$C_{th} \, dT/dt = Q_{gen} - hA_s(T - T_{amb})$$

Cell temperature  $T$  evolves from heat generation, convection, and ambient temperature.

(18) DISTRIBUTED HEAT CONDUCTION

$$\rho c_p \, \partial T / \partial t = \nabla \cdot (k_{th} \nabla T) + q_{gen}$$

Spatial thermal field equation with density  $\rho$ , heat capacity  $c_p$ , and conductivity  $k_{th}$ .

(19) ARRHENIUS REACTION RATE

$$k_{rxn}(T) = A_r \exp(-E_a / (R_g T))$$

Temperature-dependent reaction-rate constant with activation energy  $E_a$ .

## Data Scope & Verification

The NASA cycle records expose voltage, current, temperature, time, and capacity, but not every internal concentration, flux, heat-source, or reaction parameter required to claim these residuals.

## Core Physical Scope

Equations (1)–(19) are the review’s core numbered map; only measurable capacity/degradation constraints are active in our Stage 1 implementation.

# Measured model comparison under the held-out-cell protocol

Mean ± standard deviation over seeds 7, 17, and 27; B0005/B0006/B0007 train and B0018 held out.

| Model                   | SOH MAE      | SOH RMSE     | SOH MAPE      | SOH R <sup>2</sup> | RUL MAE (cycles) |
|-------------------------|--------------|--------------|---------------|--------------------|------------------|
| DNN                     | 11.05 ± 0.25 | 21.03 ± 0.08 | 13.74 ± 0.36% | -6.43              | 37.04 ± 6.18     |
| GRU                     | 10.52 ± 1.03 | 13.05 ± 0.63 | 13.69 ± 1.46% | -1.87              | 36.39 ± 4.95     |
| Transformer-only        | 11.50 ± 0.56 | 21.22 ± 0.31 | 14.42 ± 0.83% | -6.57              | 43.98 ± 9.94     |
| PINN without adaptation | 7.76 ± 1.42  | 15.81 ± 6.27 | 9.65 ± 1.53%  | -3.86              | 38.15 ± 4.59     |
| Adaptive twin           | 10.05 ± 1.46 | 17.22 ± 5.45 | 12.82 ± 1.78% | -4.48              | 38.36 ± 3.39     |

Readout

Standalone PINN has the lowest mean frozen SOH MAE (7.76 ± 1.42). The adaptive twin’s value is the later-stream improvement and low update cost, not a claim of the best frozen-cell score.

# Adaptation targets, ablations, and robustness results

All required experiment families were executed; the acceptance table is the direct proof-of-concept decision record.

| Acceptance measure      | Mean ± SD      | Target | Status |
|-------------------------|----------------|--------|--------|
| Post-shift improvement  | 80.63 ± 2.46%  | ≥ 10%  | PASS   |
| Adaptation work ratio   | 8.75 ± 0.00%   | ≤ 10%  | PASS   |
| Prior-domain forgetting | -10.20 ± 4.47% | ≤ 5%   | PASS*  |
| Drift events            | 7 ± 0          | > 0    | PASS   |

| Scenario        | Frozen MAE | Adapted MAE | Events |
|-----------------|------------|-------------|--------|
| Clean           | 8.81       | 1.73        | 7      |
| V/I/T noise     | 9.02       | 1.52        | 7      |
| Missing samples | 8.78       | 1.51        | 7      |
| Sensor drift    | 8.88       | 1.69        | 7      |

Ablation post-shift SOH MAE

Proposed 1.74 · No physics 1.23 · No Transformer 4.09 · No replay 3.97 · No drift detector 1.59 · Full retraining 1.58

Physical checks: SOH bounds 0.00% · negative RUL 0.00% · capacity-curve residual  $0.00573 \pm 0.00009$  · \*negative forgetting means prior-domain MAE improved

# What the base review and prior papers actually prove

The project result is new evidence from the executed NASA protocol; literature values remain context because protocols are not directly equivalent.

| Source                        | Experiment / formula focus                                                                                      | Reported result or evidence                                                                                |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 0.base_paper.pdf              | Methodological review: PINN residuals, electrochemical/ECM/aging/thermal equations, uncertainty, deployment     | No single new end-to-end model or one reproducible score; it synthesizes reviewed methods.                 |
| Applied Energy paper          | Multi-task PINN + simplified Transformer; CV charging features; sensitivity and transfer learning               | SOH MAPE 0.75%; degradation-path deviation $\approx 0.01$ ; RUL MAE 104 cycles in reported cases.          |
| Energy EM-PINN                | SEI-layer-growth PDE with PDE, BC, IC, and SOH losses; XJTU/MIT/HUST; six data-volume scenarios                 | Reported MAPE as low as 0.873%.                                                                            |
| Scientific Reports Bat-T-GNN  | Cycle-aware patches, time-aware convolution, Transformer, dynamic GNN, PINN-RUL; ablations and corruption tests | Reported NASA values include RMSE 0.014, MAE 0.007, MAPE 0.008; robustness evidence is central.            |
| Complex & Intelligent Systems | PINN + physics-guided augmentation + momentum contrastive learning on B0005/B0006/B0007/B0018                   | Compared CNN, BPINN, Informer, XGBoost-ARIMA; added ablation, SHAP, parameter, time, and latency analysis. |

Key Contribution

Our core contribution is the integration of drift-triggered replay adaptation with physics-preserving constraints for real-time BMS deployment.